

Bits 7:4 **HPRE[3:0]**: AHB prescaler

Set and cleared by software to control the division factor of the AHB clock.

Caution: Depending on the device voltage range, the software must set correctly these bits to ensure that the system frequency does not exceed the maximum allowed frequency (for more details, refer to [Section 8.2.5: Dynamic voltage scaling management](#)). After a write operation to these bits and before decreasing the voltage range, this register must be read to be sure that the new value has been taken into account.

0xxx: SYSCLK not divided
1000: SYSCLK divided by 2
1001: SYSCLK divided by 4
1010: SYSCLK divided by 8
1011: SYSCLK divided by 16
1100: SYSCLK divided by 64
1101: SYSCLK divided by 128
1110: SYSCLK divided by 256
1111: SYSCLK divided by 512

Bits 3:2 **SWS[1:0]**: system clock switch status

Set and cleared by hardware to indicate which clock source is used as system clock.

00: MSI oscillator used as system clock
01: HSI16 oscillator used as system clock
10: HSE used as system clock
11: PLL used as system clock

Bits 1:0 **SW[1:0]**: system clock switch

Set and cleared by software to select system clock source (SYSCLK).

Configured by hardware to force MSI oscillator selection when exiting Standby or Shutdown mode. Configured by hardware to force MSI or HSI16 oscillator selection when exiting Stop mode or in case of failure of the HSE oscillator, depending on STOPWUCK value.

00: MSI selected as system clock
01: HSI16 selected as system clock
10: HSE selected as system clock
11: PLL selected as system clock